rexroth
A Bosch Company

Quick Start Guide

ctrlX DRIVE

Technology Function

First Steps

R911409766 Edition 05

Disclaimer

The data specified above only serve to describe the product. As our products are constantly being further developed, no statements concerning a certain condition or suitability for a certain application can be derived from our information. The information given does not release the user from the obligation of own judgment and verification. It must be remembered that our products are subject to a natural process of wear and aging.

DOK-XDRV**-TECHFUNC***-QU05-EN-P

DC-AE (sa), (ka); DC/ESW-FA1 (bb)

38ce04441289c6630a347e886f938c05, 5, en_US

Table of contents

1 About this documentation

This documentation explains how to install the Engineering and programming tools, how to activate the "Technology Function" and how to create a simple PLC program for ctrlX DRIVE.

Editions of this documentation

Edition	Release date	Comment
01	2021-03-05	First edition
02	2021-07-13	Modifications: • In ➦ Chapter 4 Enabling the functional package "ctrlX DRIVE Technology Function" on page 11, step 2 contained a prompt to select a project. However, the connection to a device has to be established in step 2. • In ➦ Chapter 7 Establishing the communication to the ctrlX DRIVE target on page 14, step 5 was added.
03	2022-07-04	Modifications: • In ➦ Chapter 4 Enabling the functional package "ctrlX DRIVE Technology Function" on page 11, step 5 was removed ("Activate selection"). After a functional package was selected, the selection does not need to be activated anymore. • More detailed information included in ➦ Chapter 8 Axis control on page 14.
04	2024-01-04	• Development of individual PLC programs from AXS-V-0316 for ctrlX DRIVE possible with license "T0" • Package installation with ctrlX PLC Engineering described from version 1.18.x • Described subsequent installation of libraries
05	2024-09-13	Updated "➦ Chapter 2 Installing Engineering tools on the PC on page 4"

Feedback on this documentation

Your experience is an important part of the product and documentation improvement process.

In case of any errors or if you want to suggest changes to this documentation, please do not hesitate to contact us.

Please send your feedback to ➦ dokusupport@boschrexroth.de.

2 Installing Engineering tools on the PC

This section describes how to install the Engineering tools required for using "Technology Function". The following software is required:

- ctrlX WORKS +
- ctrlX DRIVE Engineering +
- ctrlX PLC Engineering +

Installing ctrlX WORKS

Proceed as follows when installing ctrlX WORKS:

1. Download the installation files of ctrlX WORKS (see ➦ "FAQ for ctrlX WORKS", "Where can I get ctrlX WORKS?").
2. To start the installation, execute the setup file (“ctrlx-works-xxxx.exe”) of ctrlX WORKS (admin rights required).
   - ➡ The user account control is displayed and you are asked whether to allow the "PackageManager" to be executed.
3. Allow the "PackageManager" to be executed.
   - ➡ The welcome screen of the ctrlX WORKS installation is displayed.
4. Select the installation language and confirm the dialog with “Next”.
   - ➡ The terms of use of Bosch Rexroth AG are displayed.
5. Read the terms of use. To confirm the terms of use, start the installation with “Accept”.
   - ➡ The dialog for selecting the installation target directory is opened.
6. Specify the directory for the ctrlX WORKS installation and confirm the dialog with “Next”.
   - ➡ The dialog for selecting the installation options is displayed.
7. Using "Technology Function" only requires the ctrlX WORKS option. You may deselect all other options.

   Confirm your selection with “Next”.
   - ➡ A list of the required software packages is identified and displayed.
8. Start the installation with “Install”
   - ➡ The installation is started; it may take some minutes and the installation progress is displayed visually.

     The installation result is displayed.
9. To complete the installation, select “Finish”.

Installing the Engineering tools

Proceed as follows to install the Engineering tools:

1. Start ctrlX WORKS.
2. On the sidebar click “Settings”
   - ➡ The “Settings” window opens. This window contains the settings for ctrlX WORKS.
3. Using "Technology Function" requires the Engineering tools ctrlX DRIVE Engineering and ctrlX PLC Engineering.

   Click the “Setup Engineering Tools” tile.
   - ➡ The “Setup Engineering Tools” window opens. This window displays the Engineering tools already installed and those available for installation. If required, Engineering tools already installed may be updated in this window, insofar as a new version of these tools is available for installation. For the initial installation of ctrlX DRIVE Engineering click the corresponding tile, and in the following view click “Install”.
   - ➡ The user account control is displayed and you are asked whether to allow the "PackageManager" to be executed.
4. Allow the "PackageManager" to be executed.
   - ➡ The welcome screen of the ctrlX DRIVE Engineering installation is displayed.

     Note: By default, the "Run setups in the background" option is active and ensures that only the query for the "administrator mode" has to be confirmed when a setup is run. Everything else is run in the background. (Does not apply to the initial installation of a tool. In this case, it is required to select an installation directory.). Only if the "Run setups in the background" option has been deactivated, will the user interface of the tool setup (steps 5-10) appear with every update.
5. Select the installation language and confirm the dialog with “Next”.
   - ➡ The terms of use of Bosch Rexroth AG are displayed.
6. Read the terms of use. To confirm the terms of use, start the installation with “Accept”.
   - ➡ The dialog for selecting the installation target directory is opened.
7. Specify the directory for the ctrlX DRIVE Engineering installation and confirm the dialog with “Next”.
   - ➡ The dialog for selecting the installation options is displayed.
8. In the “Functions / ctrlX DRIVE system” section, select the “Commissioning and diagnostics” option, and in “Functions / Service”, select the “Device data sheets for ctrlX DRIVE” option. Other options are not required for using "Technology Function". In the “Languages” section, you may additionally select the languages to be installed. Confirm your selection with “Next”.
   - ➡ A list of the required software packages is identified and displayed.

9. Start the installation with “Install”.
   ➡ The installation is started; it may take some minutes and the installation progress is shown visually. In case ctrlX DRIVE Engineering is installed for first time, a query regarding the installation of the device software "Robert Bosch GmbH Network Adapter" is displayed during the installation. If you use ctrlX DRIVE with ctrlX DRIVE Panel and would like to use it to connect the PC to the drive controller, confirm the installation of the device software with “Install”. The installation result is displayed.
10. To complete the installation, select “Finish”.
11. Click the tile of ctrlX PLC Engineering and in the following view click “Install”.
    ➡ The user account control is displayed and you are asked whether to allow the "PackageManager" to be executed.
12. Allow the "PackageManager" to be executed.
    ➡ The welcome screen of the ctrlX PLC Engineering installation is displayed.

    Note: By default, the "Run setups in the background" option is active and ensures that only the query for the "administrator mode" has to be confirmed when a setup is run. Everything else is run in the background. (Does not apply to the initial installation of a tool. In this case, it is required to select an installation directory.). Only if the "Run setups in the background" option has been deactivated, will the user interface of the tool setup (steps 13-18) appear with every update.
13. Select the installation language and confirm the dialog with “Next”.
    ➡ The terms of use of Bosch Rexroth AG are displayed.
14. Read the terms of use. To confirm the terms of use, start the installation with „Accept“.
    ➡ The dialog for selecting the installation target directory is opened.
15. Specify the directory for the ctrlX PLC Engineering installation and confirm the dialog with “Next”.
    ➡ The dialog for selecting the installation options is displayed.
16. The selection of the ctrlX PLC Engineering option is mandatory. Confirm your selection with “Next”.
    ➡ A list of the required software packages is identified and displayed.
17. Start the installation with “Install”.
    ➡ The installation is started; it may take some minutes and the installation progress is shown visually. The installation result is displayed.
18. To complete the installation, select “Finish”.

3 Package installation ctrlX DRIVE

To program a Technology App on ctrlX DRIVE, a package has to be installed in ctrlX PLC Engineering. With the installation of the package, the device description and the required libraries are installed.

This section describes how to install the package; 2 different cases are described:

- Installing the package with ctrlX PLC Engineering version 1.16.x and below
- Installing the package with ctrlX PLC Engineering version 1.18.x and above

Installing the package with ctrlX PLC Engineering version 1.16.x and below

Requirements

- ctrlX PLC Engineering (≤ version 1.16.x) must have been installed.
- The package *ctrlX_DRIVE_TF_Vx.x.x.x.package* has to be available on the PC.

1. Start ctrlX PLC Engineering.
2. Open the *“Tools → Package Manager...”* menu
   ➡ The “Package Manager” dialog is opened.
3. Click the “Install...” button
   ➡ The dialog for selecting the package is opened.
4. Navigate to the directory that contains the package *ctrlX_DRIVE_TF_Vx.x.x.x.package*, and select the package. Click the “Open...” button.
   ➡ The “Choose Setup Type” dialog is opened.
5. Select the option “Typical setup”.
   ➡ The standard set of device description file and libraries defined in the package is installed.

Installing the package withctrlX PLC Engineering version 1.18.x and above

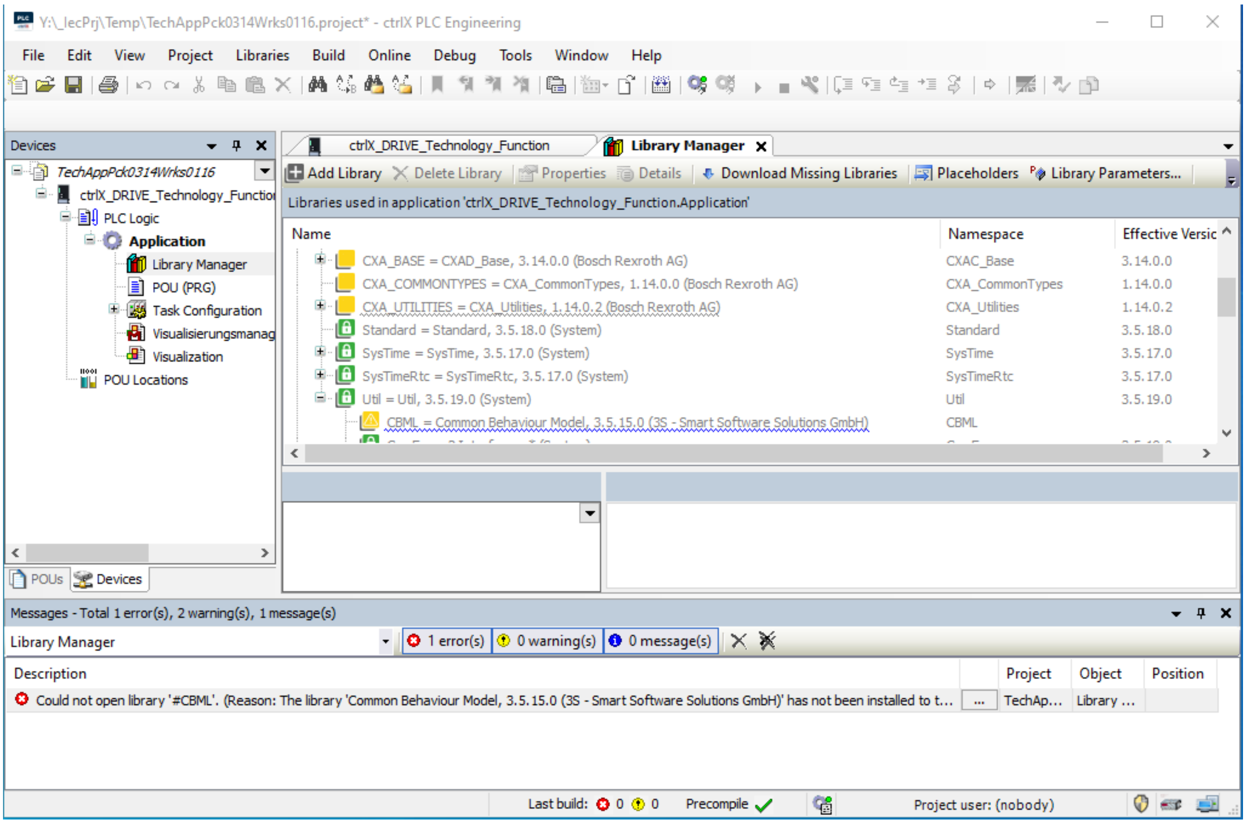
Requirements

- ctrlX PLC Engineering (≥ version 1.18.x) must have been installed.
- The package *ctrlX_DRIVE_TF_Vx.x.x.x.package* has to be available on the PC.

1. Start ctrlX PLC Engineering.
2. Open the *“Tools → Add-on Installer”* menu
3. Allow the "User Account Control" to start the "APInstaller.GUI" program.
   ➡ The “Add-on Installer” dialog is opened.
4. Click the “Install file” button
   ➡ The dialog for selecting the package is opened.
5. Navigate to the directory that contains the package *ctrlX_DRIVE_TF_Vx.x.x.x.package*, and select the package. Click the “Open” button.
   ➡ The confirmation dialog is opened. Confirm the installation with the “OK” button.
6. If the package ctrlX DRIVE Technology Function does not yet have a signature, you have to confirm the missing signature by ticking the check box “I want to continue despite the missing signature(s)”. Click the “Next” button.
   ➡ The installation of the package *ctrlX_DRIVE_TF_Vx.x.x.x.package* is started, and if necessary you will be requested in a message box to close ctrlX PLC Engineering.
7. Close ctrlX PLC Engineering.
8. Confirm the message window with the “OK” button.
   ➡ The standard set of device description file and libraries defined in the package is installed.
9. After successful installation, you can restart ctrlX PLC Engineering.

Downloading missing libraries

When an older project is edited with an updated version of ctrlX PLC Engineering or after an update of ctrlX PLC Engineering, required library versions might not have been installed. If required library versions have not been installed, this is signaled as an error during compilation and displayed in the library manager:



In this case, the missing libraries can be subsequently installed in the library manager via the Internet with the “Download Missing Libraries” command. Depending on the configuration of the company network, it might be required to create and use proxy settings [*“Tools → Options → Proxy settings”* (see also ctrlX PLC Engineering, Application Manual)].

4 Enabling the functional package "ctrlX DRIVE Technology Function"

This section describes how to enable the functional package ctrlX DRIVE Technology Function in ctrlX DRIVE Engineering.

Prerequisite

- "Technology Function" "TE1", "TE0" or "TX1" must have been licensed.

  NOTE: "Technology Function" "TF1" is not sufficient for programming ctrlX DRIVE Technology Function!

  If the device was not ordered with the required license, please contact our service department to have the "Technology Function" "TE1", "TE0" or "TX1" subsequently licensed.

1. Start ctrlX DRIVE Engineering.
2. Establish the connection to a device fulfilling the specified requirement.
3. Call the menu *"Commissioning → Enabling firmware functions"*.

   ➡ The "Enabling firmware functions" dialog is opened.
4. Tick the check box of the firmware function ctrlX DRIVE Technology Function ("Technology Function" section).
5. Restart the device by switching it off and back on.

5 Creating a project for ctrlX DRIVE

This section describes the prerequisites and steps required to create a project for ctrlX DRIVE in ctrlX PLC Engineering.

Prerequisites

- ctrlX PLC Engineering must have been installed.
- The package *ctrlX_DRIVE_TF_Vx.x.x.x.package* has to be available.

1. Start ctrlX PLC Engineering.
2. Press the shortcut [ctrl]+[n] or select *“File → New Project...”* in the menu.
   ➡ A dialog to create a new project is opened.
3. Select the template “ctrlX DRIVE Technology Function”.
4. To confirm the selection, click “OK”.
   ➡ A project with "ctrlX DRIVE Technology Function" as the device is created. The program function block "PLC_PRG" with the programming language "Structured Text (ST)" is created.

Alternatively, a new project can be created using the “Standard project” template. For this purpose, select "ctrlX DRIVE Technology Function (Bosch Rexroth AG)" as the device.

6 Example programs for ctrlX DRIVE

Example programs for the function blocks are available in ctrlX PLC Engineering in the library manager:

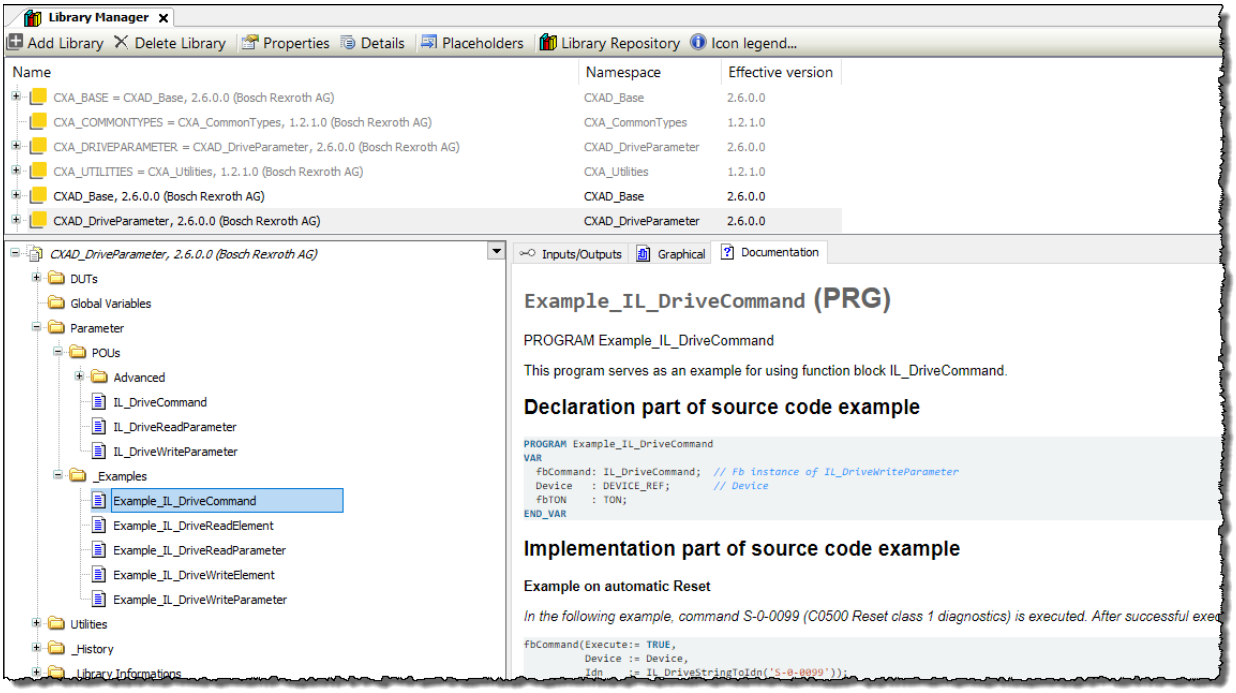


Fig. 1: Example programs for the POUs IL_DriveReadElement, IL_DriveWriteElement, IL_DriveCommand, IL_DriveReadParameter and IL_DriveWriteParameter

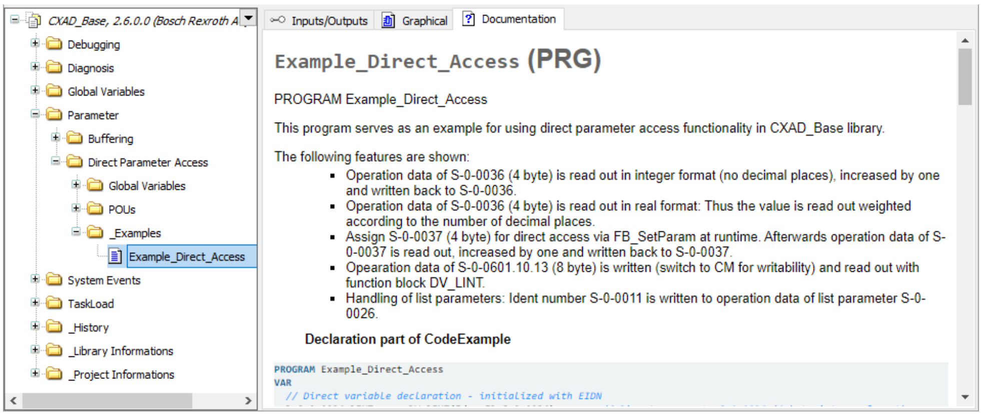


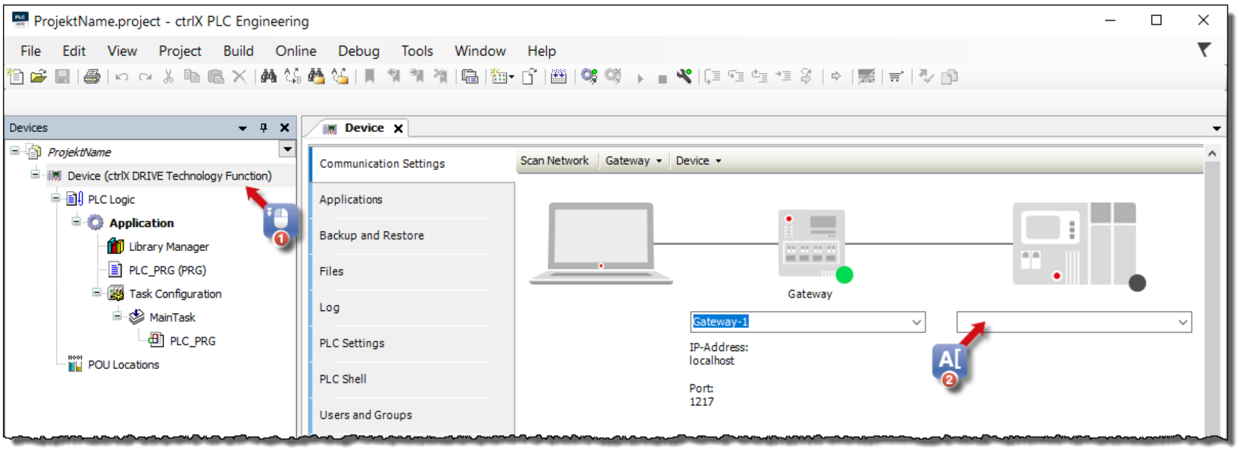
Fig. 2: Example program for acyclic access to axis parameters via direct variables

7 Establishing the communication to the ctrlX DRIVE target

This section describes the prerequisites and steps required to establish the communication to the ctrlX DRIVE target in ctrlX PLC Engineering.

Prerequisites: A project for the ctrlX DRIVE target must have been created in ctrlX PLC Engineering.

1. Start ctrlX PLC Engineering.
2. Open the project created for the ctrlX DRIVE target.
3. In the “Devices” view, double-click "Device (ctrlX Drive Technology Function)" (1)
   ➡ The “Device”, “Communication Settings” window opens.



4. Enter the IP address of the axis (2) and confirm your entry with [↵].
   ➡ If the connection to the target has been established successfully, the LED at the device symbol turns green in the “Device”, “Communication Settings” window, and the target information is displayed.
5. Download the application code to the ctrlX DRIVE target system. Log in and start the PLC (refer to "➥ctrlX PLC Engineering, Application Manual")

8 Axis control

8.1 Libraries for axis control

The CXAD_Base and CXAD_AxisControl libraries contain function blocks for axis control for the target ctrlX DRIVE Technology Function.

- CXAD_AxisControl.library
  - MX_VelocityControl()
  - MX_TorqueControl()
  - MX_PositionControl()
- CXAD_Base.library
  - MX_SetControl()
  - MX_HaltBit()
  - MX_Power()
  - MX_GetExternalControl()
  - MX_PresetOpMode()

Distinguishing the ctrlX DRIVE Technology Function to the ctrlX CORE

ctrlX DRIVE Technology Function supports functionalities used to operate the drive as an "intelligent servo axis". It does not support a versatile, stand-alone single-axis "Motion Logic Control"; this purpose requires a ctrlX CORE control.

8.2 Basic principles of axis control with the integrated PLC (ctrlX DRIVE Technology Function)

The operating state of the axis is defined by external and internal control signals (error, power, etc.). The primary external control information for commanding the axis via the master control word (P‒0‒0116) is as follows:

- "Drive On"
- "Drive enable"
- "Drive Halt"
- Command operation mode input

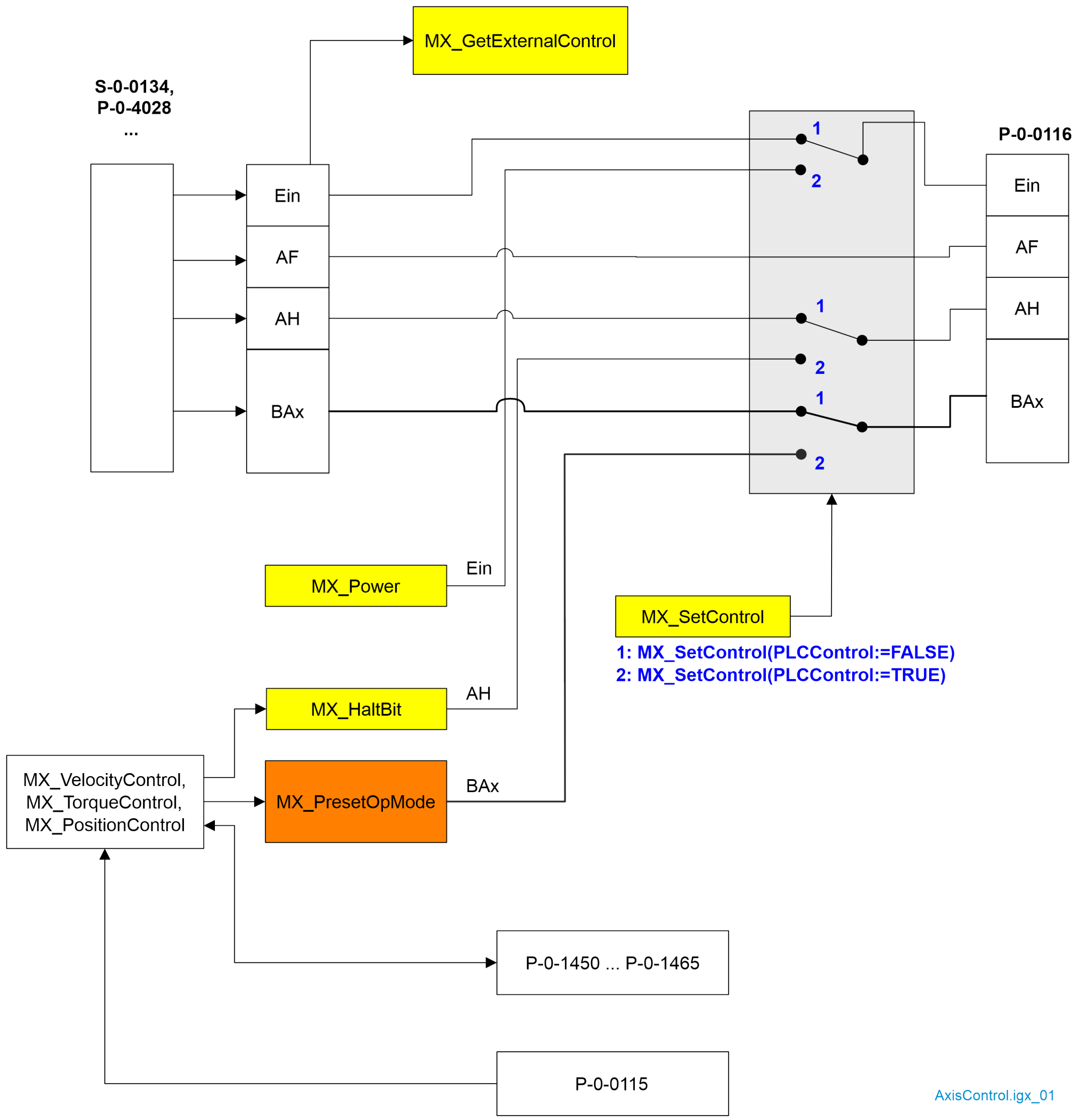
There are also drive control commands for activating complex, pre-configured commands, such as "drive-controlled homing procedure", "auto-adjustment functions", etc. Depending on the application, the external control signals are preset by an external control via master communication or by the integrated PLC (ctrlX DRIVE Technology Function).

The external control as a standard has control over the axis. The external control signals are specified via the specific control word of each master communication interface. If required, the PLC can temporarily give itself axis control using the function block MX_SetControl(). This may be necessary, e.g., for an intelligent, decentralized error reaction. This means PLC is able to temporarily control the axis via function blocks.

8.3 Commanding the axis

For commanding via the control word of the axis (P‒0‒0116) with the PLC, temporary axis control has to be activated with MX_SetControl(). Afterwards, motion control of the axis with the PLC is possible.

The parameter S‒0‒0240, bit 0 (DC Bus Power Control) is not automatically written with values, but this has to be done via a master communication or the internal PLC.



S-0-0134, P-0-4028,...	Control word of the master communication	AF	Drive enable (P-0-0116, bit 14)
P–0–0115	Status word of the axis	AH	Drive Halt (P-0-0116, bit 13)
P–0–0116	Control word of the axis		
P–0–1450..P–0–1465	Parameters of the PLC secondary modes	BAx	Target operation mode (P–0–0116, bit 8, 9, 11, 12)
Ein	Drive On (P–0–0116, bit 15)		

Controlling the axis

The PLC can get control over the axis with the function block MX_SetControl(). With the function blocks MX_Power() and MX_HaltBit(), the signals "Drive On" or "Drive Halt" (in P–0–0116) can be specified. But the external "Drive enable" signal always takes effect and is used for the external control to decelerate the drive, even in the case of temporary control.

Axis motions

Axis motions are carried out by controlling the operation modes of the drive.

The axis motions can be performed on a higher level of abstraction with easily operated function blocks of the CXAD_AxisControl library, or alternatively by activating the respective operation mode with MX_PresetOpMode() and writing the command value parameters belonging to the operation mode with direct variables. The latter, however, requires more programming effort but involves higher flexibility.

High-level programming level with function blocks of CXAD_AxisControl library

The function blocks MX_PositionControl(), MX_VelocityControl() and MX_TorqueControl() contained in the CXAD_AxisControl library are used to utilize the most important operation modes of the drive. They automatically activate the corresponding internal secondary PLC operation mode using the function block MX_PresetOpMode() and preset the command values of the function block inputs. If the drive still is in the "Drive Halt" state, this state can be canceled using the function block MX_HaltBit().

All function blocks of the CXAD_AxisControl library use the internal secondary PLC operation modes "Torque/force control", "velocity control" or "drive-controlled positioning" as independent instances of the equivalent operation modes with the parameters P–0–1450 to P–0–1465.

While temporary control is active, the parameters P–0–1450 to P–0–1465 cannot be written externally. This ensures that with temporary control, the operation mode settings of the external control are not illegally manipulated.

The CXAD_AxisControl library is available freely readable with all function blocks, so that it is possible to implement one's own function blocks with a differing or advanced functionality according to the example of the existing function blocks.

It is not allowed, however, to modify the existing function blocks! If required, the existing function blocks may be used though as templates for one's own differing function blocks.

Low-level programming level with MX_PresetOpMode() and parameter access

For specific tasks, that are not covered by the ready-made function blocks for controlling the operation modes, it is alternatively possible to directly activate any desired operation mode using the function block MX_PresetOpMode() and to preset all required parameters via direct variables.

9 Simulating an axis

If a motor is not available, an axis can be operated in simulation. The following settings are required:

- ctrlX DRIVE Engineering, function tree *“<Axis> → Optimization / commissioning → Axis simulation”*
  Activating axis simulation: P-0-0399, bit 0="1"
- ctrlX DRIVE Engineering, function tree *“<Axis> → Motor, drive mechanics, measuring systems → Encoder1 / motor encoder → Encoder1 / motor encoder”*
  Setting evaluation of encoder interface "Option 2" (XG21) for encoder 1: P-0-0077="2"
  Deactivating absolute evaluation for encoder 1: S-0-0277, bit 7="1"

10 Documentations

10.1 Drive systems, system components

Table 1: Documentations – Drive systems, System Components

Title	Type of documentation	Document type 1)	Material number
ctrlX DRIVE Drive Systems	Project Planning Manual	DOK-XDRV**-X***********-PRxx-EN-P	R911386579
ctrlX DRIVE DC/DC Converter XMV	Application Manual	DOK-XDRV**-XMV********-APxx-EN-P	R911413650
Security Instructions Electric Drives and Controls	Project Planning Manual	DOK-IWORKS-SECURITY***-PRxx-EN-P	R911342562
Control Cabinet Air Conditioning, EMC, Design, IP Code, IndraDrive Electrics, Rexroth EFC/Fv, Sytronix	Project Planning Manual	DOK-DRIVE*-CABINET****-PRxx-EN-P	R911344988

1) In the document type codes, "xx" is a placeholder for the current edition of the documentation (e.g.: PR01 is the first edition of a Project Planning Manual)

10.2 Firmware/Runtime

Table 2: Documentations – firmware

Title	Type of documentation	Document type 1)	Material number
ctrlX DRIVE AXS-V-06 Functions	Application Manual	DOK-XDRV**-AXS-06VRS**-APxx-EN-P	R911425358
ctrlX DRIVE AXS-V-06 (CoE) Functions	Application Manual	DOK-XDRV**-AXS-06VRS*C-APxx-EN-P	R911425360
ctrlX DRIVE / eLION Diagnostic Messages of Runtime A*S-V-06RS	Reference Book	DOK-XDRV**-GEN6-DIAG**-RExx-EN-P	R911424817 2)
ctrlX DRIVE / eLION Parameters/Objects of Runtime A*S-V-06RS	Reference Book	DOK-XDRV**-GEN6-PARA*C-RExx-EN-P	R911425449 2)
ctrlX DRIVE AXS-V-05 Functions	Application Manual	DOK-XDRV**-AXS-05VRS**-APxx-EN-P	R911422255
ctrlX DRIVE AXS-V-05 (CoE) Functions	Application Manual	DOK-XDRV**-AXS-05VRS*C-APxx-EN-P	R911422257
ctrlX DRIVE Diagnostic Messages of Runtime AXS-V-05RS	Reference Book	DOK-XDRV**-GEN5-DIAG**-RExx-EN-P	R911422251 2)
ctrlX DRIVE Parameters/Objects of Runtime AXS-V-05RS	Reference Book	DOK-XDRV**-GEN5-PARA*C-RExx-EN-P	R911422253 2)

Title	Type of documentation	Document type [1)]	Material number
ctrlX DRIVE AXS-V-04 Functions	Application Manual	DOK-XDRV**-AXS-04VRS**-APxx-EN-P	R911421281
ctrlX DRIVE AXS-V-04 (CoE) Functions	Application Manual	DOK-XDRV**-AXS-04VRS*C-APxx-EN-P	R911421283
ctrlX DRIVE Diagnostic Messages of Runtime AXS-V-04RS	Reference Book	DOK-XDRV**-GEN4-DIAG**-RExx-EN-P	R911421277[2)]
ctrlX DRIVE Parameters/Objects of Runtime AXS-V-04RS	Reference Book	DOK-XDRV**-GEN4-PARA*C-RExx-EN-P	R911421279
ctrlX DRIVE AXS-V-03 Functions	Application Manual	DOK-XDRV**-AXS-03VRS**-APxx-EN-P	R911410073
ctrlX DRIVE AXS-V-03 (CoE) Functions	Application Manual	DOK-XDRV**-AXS-03VRS*C-APxx-EN-P	R911398021
ctrlX DRIVE Diagnostic Messages of Runtime AXS-V-03RS	Reference Book	DOK-XDRV**-GEN3-DIAG**-RExx-EN-P	R911409763[2)]
ctrlX DRIVE Parameters of Runtime AXS-V-03RS	Reference Book	DOK-XDRV**-GEN3-PARA**-RExx-EN-P	R911409808[2)]
ctrlX DRIVE Parameters/Objects of Runtime AXS-V-03RS	Reference Book	DOK-XDRV**-GEN3-PARA*C-RExx-EN-P	R911419643[2)]
ctrlX DRIVE AXS-V-02 Functions	Application Manual	DOK-XDRV**-AXS-02VRS**-APxx-EN-P	R911398021
ctrlX DRIVE Diagnostic Messages of Runtime AXS-V-02RS	Reference Book	DOK-XDRV**-GEN2-DIAG**-RExx-EN-P	R911383776
ctrlX DRIVE Parameters of Runtime AXS-V-02RS	Reference Book	DOK-XDRV**-GEN2-PARA**-RExx-EN-P	R911383778

Legend:

1: *In the document typecodes, xx is a placeholder for the current edition of the documentation (e.g.: RE02 is the second edition of a Reference Book)*

2: *This documentation is only available as an online help (on "ctrlX AUTOMATION – Product Information") and not in the PDF format*

10.3 Functional safety

Table 3: Documentations – functional safety

Title	Type of documentation	Document type 1)	Material number
ctrlX SAFETY "STO" (S-/P-Parameters) ctrlX DRIVE "2nd Generation"	Application Manual	DOK-XDRV**-SI-TX******-APxx-EN-P	➦R911383774
ctrlX SAFETY "STO" (CAN Parameters) ctrlX DRIVE "2nd Generation"	Application Manual	DOK-XDRV**-SI-TX***C**-APxx-EN-P	➦R911426739
ctrlX SAFETY "SafeMotion" (S-/P-Parameters) ctrlX DRIVEplus	Application Manual	DOK-XDRV**-SI-MX******-APxx-EN-P	➦R911404905
ctrlX SAFETY "SafeMotion" (CAN Parameters) ctrlX DRIVEplus	Application Manual	DOK-XDRV**-SI-MX***C**-APxx-EN-P	➦R911426737

1) In the document typecodes, xx is a placeholder for the current edition of the documentation (e.g.: AP02 is the second edition of an Application Manual)

10.4 Motors

Table 4: Documentations – motors

Title	Type of documentation	Document typecode1)	Material number
MS2N Synchronous Servomotors	Project Planning Manual	DOK-MOTOR*-MS2N*******-PRxx-EN-P	➦R911347583
MS2S Synchronous Servomotors	Project Planning Manual	DOK-MOTOR*-MS2S*******-PRxx-EN-P	➦R911410075
MS2E Synchronous Servomotors acc. to ATEX Directive 2014/34/EU	Project Planning Manual	DOK-MOTOR*-MS2E*******-PRxx-EN-P	➦R911394140
MSK Synchronous Servomotors	Project Planning Manual	DOK-MOTOR*-MSK********-PRxx-EN-P	➦R911296289
MSK Synchronous Servomotors for Potentially Explosive Areas	Project Planning Manual	DOK-MOTOR*-MSK*EXGIIK3-PRxx-EN-P	➦R911312709
MKE Synchronous Motors Synchronous Servomotors acc. to ATEX Directive 2014/34/EU	Project Planning Manual	DOK-MOTOR*-MKE*GEN3***-PRxx-EN-P	➦R911411017
MAD / MAF Asynchronous Motors MAD / MAF	Project Planning Manual	DOK-MOTOR*-MAD/MAF****-PRxx-EN-P	➦R911295781
MLF Synchronous Linear Motors	Project Planning Manual	DOK-MOTOR*-MLF********-PRxx-EN-P	➦R911293635
ML3 Self-Cooled Linear Motors	Project Planning Manual	DOK-MOTOR*-ML3********-PRxx-EN-P	➦R911389760
MCL Ironless Linear Motors MCL	Project Planning Manual	DOK-MOTOR*-MCL********-PRxx-EN-P	➦R911330592

1) In the document type codes, "xx" is a placeholder for the current edition of the documentation (e.g.: PR01 is the first edition of a Project Planning Manual)

10.5 Cables

Table 5: Documentations – Cables

Title	Type of documentation	Document type [1)]	Material number
ctrlX Motor Cables and Connectors	Reference Book	DOK-CONNEC-XDRV*******-RExx-EN-P	➦R911420100
Motor cables and connections with IndraDrive	Product information	DOK-CONNEC-MS2N*INDRV*-CAxx-EN-P	➦R911401938
Rexroth Connection Cables IndraDrive and IndraDyn	Selection Data	DOK-CONNEC-CABLE*INDRV-CAxx-EN-P	➦R911322949

1) In the document type codes, xx is a placeholder for the current edition of the documentation (e.g.: CA03 is the third edition of the Catalog documentation)

11 Service and support

Our worldwide service network provides an optimized and efficient support. Our experts provide you with advice and assistance. You can contact us **24/7 - including weekends and public holidays**.

Service Germany

Our technology-oriented Competence Center in Lohr, Germany, is responsible for all your service-related queries for electric drive and controls.

Contact the **Service Hotline** and **Service Helpdesk** under:

Tel:	+49 9352 40 5060
Fax:	+49 9352 18 4941
Email:	➦service.svc@boschrexroth.de
Internet:	➦http://www.boschrexroth.com

Additional information on service, repair (e.g. delivery addresses) and training can be found on our internet sites.

Service worldwide

Outside Germany, please contact your local service office first. For hotline numbers, refer to the sales office addresses on the internet.

Preparing information

To be able to help you more quickly and efficiently, please have the following information ready:

- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)

Furthermore, please backup all parameters or generate a system report in ctrlX DRIVE Engineering (“Help”→“Generate system report”).

12 Glossary

Application Manual
The application manual comprises the entire documentation which is used to provide information to the user of the product about the use and the safety-relevant contents for project planning, assembly, installation, mounting, commissioning, operation, maintenance, repairs and decommissioning of the product. The following terms are used for the application manual: Operating Instructions, Commissioning Manual, Instruction Manual, Project Planning Manual, Application Manual, etc.

Component
A component is a combination of assembly parts with a specified function which are part of the equipment, the device or the system. Components of the electric drive and control system are e.g. supply units, drive control devices, mains choke, mains filter, motors, cables, etc.

ctrlX AUTOMATION
ctrlX AUTOMATION is the open and scalable automation platform; it breaks down the conventional boundaries between machine control, IT world and Internet of Things.

ctrlX CORE
ctrlX CORE is the product line of the compact control platform of ctrlX AUTOMATION.

ctrlX CORE is available in embedded form, drive-integrated form or in the IPC.

ctrlX DRIVE
ctrlX DRIVE is the product line of the compact modular drive system of ctrlX AUTOMATION.

ctrlX DRIVE Engineering
ctrlX DRIVE Engineering is the software used to configure and commission the ctrlX DRIVE drive system.

ctrlX DRIVE Technology Function
ctrlX DRIVE Technology Function is the PLC firmware function that allows customized PLC programs or ready-made Technology Apps to be used in the axis processor of the ctrlX DRIVE drive system.

ctrlX DRIVEplus
ctrlX DRIVE is the product line of the compact modular drive system of ctrlX AUTOMATION.

With ctrlX DRIVEplus, the drives can be extended by additional software functions and hardware.

ctrlX PLC Engineering
ctrlX PLC Engineering is the development environment in accordance with the IEC 61131-3 standard for programmable logic controllers of ctrlX AUTOMATION.

ctrlX SAFETY
ctrlX SAFETY is an umbrella term for the safety technology product lines of ctrlX AUTOMATION.

ctrlX SAFETY refers to compact safety controllers of the SAFEX-C.12 and SAFEX-C.15 types.

Other characteristics of safety technology product lines are ctrlX DRIVEplus and ctrlX SAFETYplus.

ctrlX WORKS
ctrlX WORKS is the central software of ctrlX AUTOMATION. In ctrlX WORKS, all available ctrlX devices in the network are visible.

Furthermore, ctrlX WORKS can be used to install and start app-based and web-based engineering and programming tools for typical automation tasks. It is possible to develop your own applications in any programming language, and third-party apps can be integrated in ctrlX WORKS.

Device
A device is an end product with an individual function, intended for the user and put on the market as individual commodity.

Drive
A drive (electric drive) consists of a drive controller with an electric motor.

Installation
An installation consists of multiple devices or systems interconnected for a defined purpose and at a defined location. However, these devices or systems are not intended to be put on the market as a single functional unit.

Package
A package is an installable artifact that contains one or more software artifacts (1..n) which can be used on a device.

A package can contain, for example, firmware artifacts, applications, templates and recipes. Packages are customized features a customer can buy and/or install.

Product
Example of a product: Device, component, part, system, software, firmware, among other things.

Project Planning Manual
A Project Planning Manual is part of the application documentation used to assist in the sizing and planning of systems, machines or installations.

User
A user is a person installing, commissioning or using a product which has been placed on the market.

13 Index

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